Showcase Portfolio

TEAM SUPREME - GROUP #9

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OVERVIEW

- Our team has created a hands-on learning tool for K-12 students to have them actively participate in engineering.
- Students will assemble, program, and hardwire a toy dump truck.
- Features of the truck:
  - Triangulated ultrasonic sensors
  - Mechanical truck bed
  - Horn

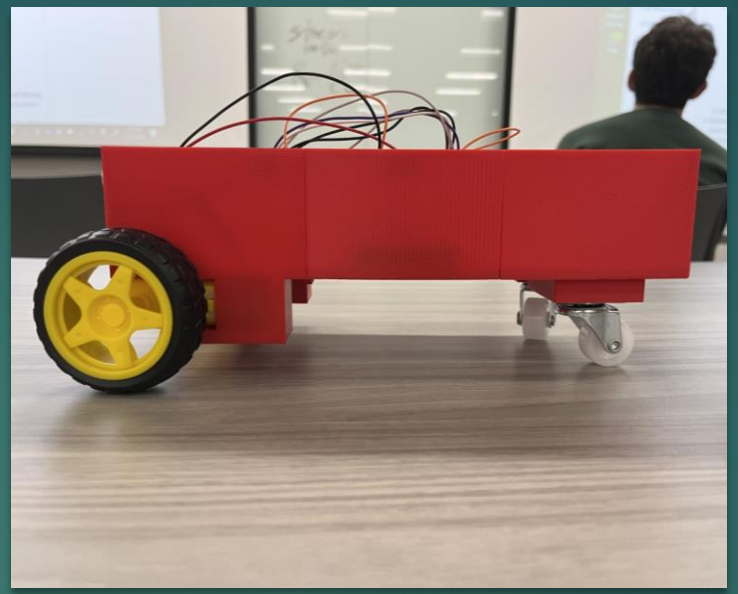
MOTIVATION

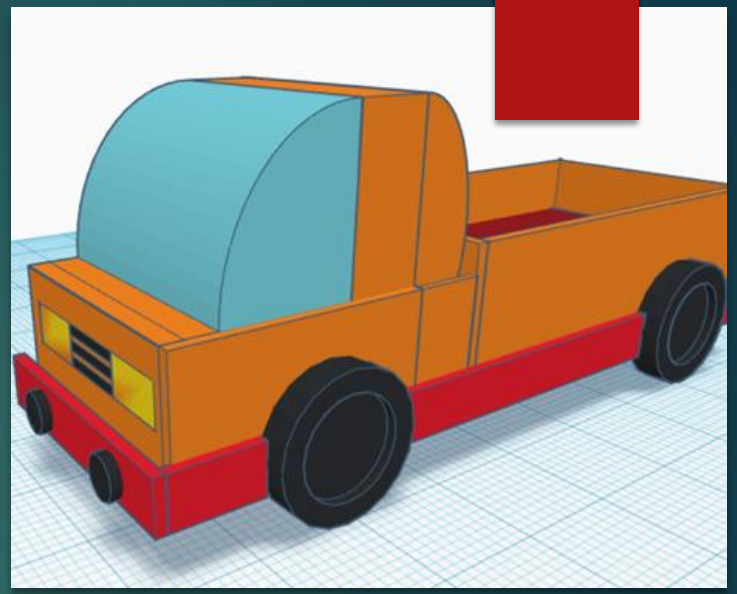
- Our motivations for creating this robot were to foster interest in engineering among K-12 students.
- We wish to:
  - Give students a hands-on experience in a fun and exciting way.
  - Provide insights into various mechanical and electrical aspects of building a toy.

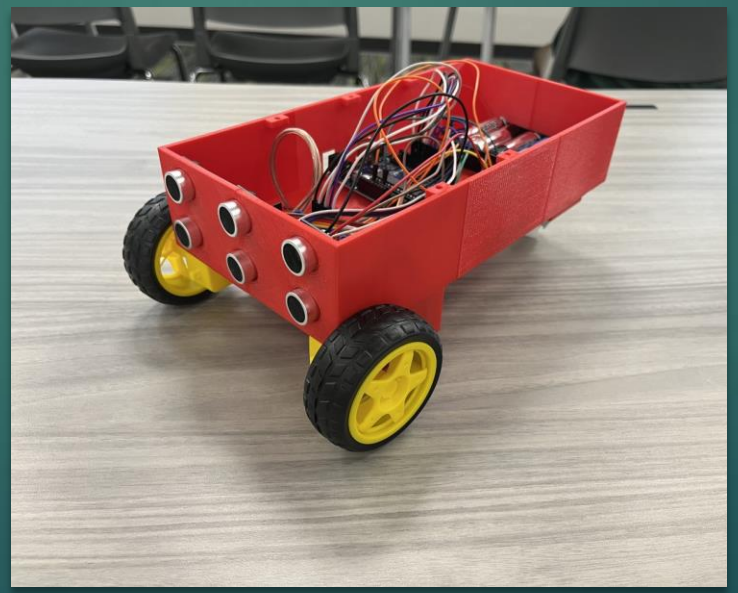
WE LOVE TO DUMP

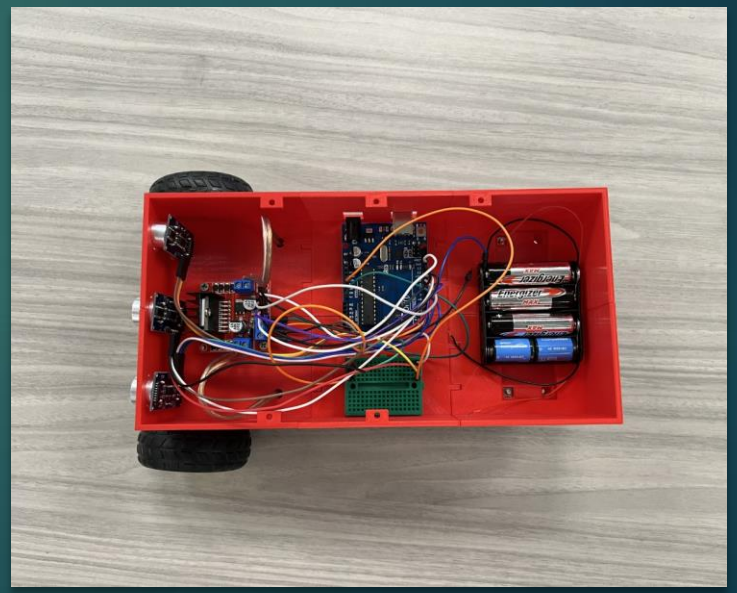
PRODUCT DEVELOPMENT

- We decided on a dump truck because it has a simple, fun design with useful features.
- Three ultra-sonic sensors are employed to control the movement and turning of the truck.
- Areas of revisions:
  - Coding- the speed was too fast causing the wheels to slip
  - Motorized truck bed- Presented a pinching hazard, gears were too fragile for the amount of torque.









FINAL DESIGN

- Dimensions: 10" x 5" x 7"
- Materials:
  - PLA/3D printed chassis
  - Construction paper
  - Recycled cardboard

Materials	Cost
Arduino Op Amp	$4.27 $0.10
Breadboard PLA or ABS Filament	$1.43 $0.02 per gram
2 pack of wheels Capacitor Kit	$3.32 $1.22
Sensor Capacitor kit	$3.40 x 3 $1.22
Battery Resistor Kit	$2.00 $1.30
Soldering wire pack	$0.50
LEDs Screws and nuts Total	$0.10 $0.09 $25.95

ENGINEERING TECHNOLOGY

```
void setup() {
  Serial.begin(38400);
  //Set Trigpins as output pins
  pinMode(trigPin1, OUTPUT);
  pinMode(trigPin2, OUTPUT);
  pinMode(trigPin3, OUTPUT);
  //Set Echopins as input pins
  pinMode(echoPin1, INPUT);
  pinMode(echoPin2, INPUT);
  pinMode(echoPin3, INPUT);
  //Set PWM pins as OUTPUT pins
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);
  //analogWrite(ENA, MotorSpeed);
  //analogWrite(ENB, MotorSpeed);
  //Include the motor speeds
//  motor1.setSpeed(Speed);
//  motor2.setSpeed(Speed);
  //Set up speaker pin
  pinMode(9, OUTPUT);
}
```

```
  if (centerSensor <= 20) {
    backward();
    tone(11,NOTE_F5,75);
    Serial.println("Backing up!");
    delay(25);
    Stop();
    delay(50);
    noTone(11);
  }
  else if (centerSensor <= 25) {
    Stop();
    Serial.println("Stop");
    delay(75);
  } else if (centerSensor <= 100) {
    forward();
    delay(50);
    Stop();
    delay(25);
  } else {
      if (leftSensor - rightSensor > 3 ) {
        right();
        Serial.println("Right");
        delay(50);
      } else if (rightSensor - leftSensor > 3) {
        left();
        Serial.println("Left");
        delay(50);
      }
      Serial.println("Stop");
      Stop();
      delay(25);
}
```

PRODUCT SPECIFICATIONS

Power	Weight	Turn Radius	Speed	Size	Storage Size	Sensing Mode	Safety (-/10)	Operation
15V (lithium ion+ alkaline)	1800g	60°	7cm/s	10"x5"x7"	5"x5"x5"	3 Ultrasonic sensors	9/10	8/10

CONCLUSION

- Economical design
- Reprogrammable and customizable
- Easy to assemble
- Utilizes recyclable materials
- Promotes interactive play

RECOMMENDATION

This is a prototype if we were to mass produce this, new challenges will likely arise.

We recommend you give us $1 million to invest in our start up.

With more time, the final design could be made more colorful and aesthetically appealing.